

**6146****BEAM PENTODE****FOR POWER AMPLIFIER, OSCILLATOR, AND MODULATOR APPLICATIONS****HIGH POWER SENSITIVITY
90 WATTS CW INPUT (ICAS) UP TO 60 MC****T-12 BULB
OCTAL 8-PIN BASE****6146
ET-T1543
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2-59****DESCRIPTION AND RATING**

The 6146 is a beam pentode designed for use as a radio-frequency power amplifier and oscillator or as an audio-frequency power amplifier and modulator in mobile or fixed equipment. Featuring high power sensitivity and high efficiency, the tube may be operated with full input up to 60 megacycles and with reduced input to 175 megacycles.

The 6146 incorporates triple base-pin connections for the cathode and beam plates to permit effective r-f grounding, and an octal base with a metal sleeve having its own base-pin terminal. The sleeve, which shields the input to the tube and isolates it from the output circuit, eliminates the necessity for external shielding.

GENERAL**ELECTRICAL****Cathode—Coated Unipotential**Heater Voltage, AC or DC. $6.3 \pm 10\%$ Volts

Heater Current. 1.25 Amperes

Direct Interelectrode Capacitances*Grid-Number 1 to Plate, maximum. 0.24 μf Input. 13.5 μf Output. 8.5 μf **MECHANICAL****Mounting Position—Any****Envelope—T-12, Glass****Base—B8-86, Large Micanol-Wafer Octal 8-Pin with Sleeve or B8-98, Large Micanol-Wafer Octal 8-Pin with External Barriers and Sleeve****Top Cap—C1-1, Small****CHARACTERISTICS, MAXIMUM RATINGS, AND
TYPICAL OPERATION****AVERAGE CHARACTERISTICS**

Plate Voltage. 200 Volts

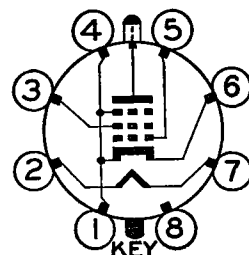
Screen Voltage. 200 Volts

Plate Current. 100 Milliamperes

Transconductance. 7000 Micromhos

Amplification Factor (Grid Number 1 to Grid Number 2). 4.5

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GENERAL  ELECTRIC**BASING DIAGRAM**

EIA 8EC

TERMINAL CONNECTIONSPin 1—Cathode, Beam Plates,
and Internal Shield

Pin 2—Heater

Pin 3—Grid-Number 2
(Screen)Pin 4—Cathode, Beam Plates,
and Internal Shield

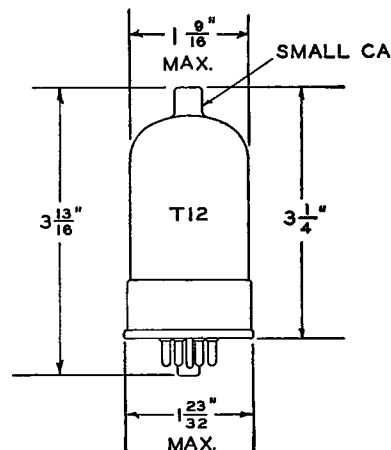
Pin 5—Grid-Number 1

Pin 6—Cathode, Beam Plates,
and Internal Shield

Pin 7—Heater

Pin 8—Base Sleeve

Cap—Plate

PHYSICAL DIMENSIONS

AUDIO-FREQUENCY POWER AMPLIFIER AND MODULATOR—CLASS AB₁+ TRIODE CONNECTION— SCREEN TIED TO PLATE

MAXIMUM RATINGS, ABSOLUTE VALUES

	CCS†	ICAS§	
DC Plate Voltage.....	400	400	Volts
Maximum-Signal DC Plate Current 	90	90	Milliamperes
Maximum-Signal Plate Input 	35	35	Watts
Plate Dissipation 	20	25	Watts
Peak Heater-Cathode Voltage			
Heater Positive with Respect to Cathode.....	135	135	Volts
Heater Negative with Respect to Cathode.....	135	135	Volts
Grid-Number 1 Circuit Resistance #			
With Fixed Bias.....	0.1	0.1	Megohms
With Cathode Bias.....	0.5	0.5	Megohms
Bulb Temperature at Hottest Point.....	220	220	C

TYPICAL OPERATION—VALUES FOR 2 TUBES

DC Plate Voltage.....	250	400	400	Volts
DC Grid-Number 1 Voltage.....	—50	—100	—100	Volts
Peak AF Grid-Number 1-to-Grid-Number 1 Voltage¶.....	100	200	200	Volts
Zero-Signal DC Plate Current.....	120	40	40	Milliamperes
Maximum-Signal DC Plate Current.....	125	100	100	Milliamperes
Effective Plate-to-Plate Load Resistance.....	5000	8000	8000	Ohms
Maximum-Signal Driving Power, approximate.....	0	0	0	Watts
Total Harmonic Distortion, approximate.....	5	4.6	4.6	Percent
Maximum-Signal Power Output, approximate.....	10	22	22	Watts

AUDIO-FREQUENCY POWER AMPLIFIER AND MODULATOR—CLASS AB₁+ TRIODE CONNECTION— SCREEN TIED TO PLATE

MAXIMUM RATINGS, ABSOLUTE VALUES

	CCS†	ICAS§	
DC Plate Voltage.....	600	750	Volts
DC Screen Voltage.....	250	250	Volts
Maximum-Signal DC Plate Current 	125	135	Milliamperes
Maximum-Signal Plate Input 	60	85	Watts
Maximum-Signal Screen Input 	3.0	3.0	Watts
Plate Dissipation.....	20	25	Watts
Peak Heater-Cathode Voltage			
Heater Positive with Respect to Cathode.....	135	135	Volts
Heater Negative with Respect to Cathode.....	135	135	Volts
Grid-Number 1 Circuit Resistance #			
With Fixed Bias.....	0.1	0.1	Megohms
With Cathode Bias.....	Not Recommended	Not Recommended	
Bulb Temperature at Hottest Point.....	220	220	C

TYPICAL OPERATION, VALUES FOR 2 TUBES

DC Plate Voltage.....	400	500	600	600	750	Volts
DC Screen Voltage.....	190	185	180	200	195	Volts
DC Grid-Number 1 Voltage						
From Fixed Bias Source.....	—40	—40	—45	—50	—50	Volts
Peak AF Grid-Number 1-to-Grid Number 1 Voltage.....	80	80	90	100	100	Volts
Zero-Signal DC Plate Current.....	63	57	26	28	23	Milliamperes
Maximum-Signal DC Plate Current.....	228	215	200	229	220	Milliamperes
Zero-Signal DC Screen Current.....	2.5	2.0	1.0	1.0	1.0	Milliamperes
Maximum-Signal DC Screen Current.....	25	25	23	27	26	Milliamperes
Effective Plate-to-Plate Load Resistance.....	4000	5500	7000	6000	8000	Ohms
Maximum-Signal Driving Power, approximate.....	0	0	0	0	0	Watts
Total Harmonic Distortion, approximate.....	8	8	8	7.5	5.7	Percent
Maximum-Signal Power Output, approximate.....	55	70	82	95	120	Watts

AUDIO-FREQUENCY POWER AMPLIFIER AND MODULATOR—CLASS AB,**
MAXIMUM RATINGS, ABSOLUTE VALUES

	CCS†	ICAS§	
DC Plate Voltage	600	750	Volts
DC Screen Voltage	250	250	Volts
Maximum-Signal DC Plate Current 	125	135	Milliamperes
Maximum-Signal Plate Input 	62.5	90	Watts
Maximum-Signal Screen Input 	3.0	3.0	Watts
Plate Dissipation	20	25	Watts
Peak Heater-Cathode Voltage			
Heater Positive with Respect to Cathode	135	135	Volts
Heater Negative with Respect to Cathode	135	135	Volts
Grid-Number 1 Circuit Resistance††			
With Fixed Bias	30000	30000	Ohms
With Cathode Bias	Not recommended	Not recommended	
Bulb Temperature at Hottest Point	220	220	C

TYPICAL OPERATION, VALUES FOR 2 TUBES

DC Plate Voltage	400	500	600	600	750	Volts
DC Screen Voltage	175	175	165	190	165	Volts
DC Grid-Number 1 Voltage						
From Fixed Bias Source	-41	-44	-44	-48	-46	Volts
Peak AF Grid-Number 1-to-Grid-Number 1 Voltage	95	102	97	109	108	Volts
Zero-Signal DC Plate Current	33	27	22	28	22	Milliamperes
Maximum-Signal DC Plate Current	232	242	207	270	240	Milliamperes
Zero-Signal DC Screen Current	1.1	0.7	0.6	1.2	0.3	Milliamperes
Maximum-Signal DC Screen Current	18	18	17	20	20	Milliamperes
Maximum-Signal DC Grid-Number 1 Current	1.6	1.9	1.1	2.0	2.6	Milliamperes
Effective Plate-to-Plate Load Resistance	3700	4600	6800	5000	7400	Ohms
Maximum-Signal Driving Power, approximate††	0.2	0.3	0.2	0.3	0.4	Watts
Total Harmonic Distortion, approximate	9.7	9.7	9.7	11	10	Percent
Maximum-Signal Power Output, approximate	62	83	90	113	131	Watts

PLATE-MODULATED RADIO-FREQUENCY POWER AMPLIFIER—CLASS C TELEPHONY CARRIER CONDI-
TIONS PER TUBE FOR USE WITH A MAXIMUM MODULATION FACTOR OF 1.0
MAXIMUM RATINGS, ABSOLUTE VALUES

	CCS†	ICAS§	
DC Plate Voltage	480	600	Volts
DC Screen Voltage	250	250	Volts
DC Grid-Number 1 Voltage	-150	-150	Volts
DC Plate Current	117	125	Milliamperes
DC Grid-Number 1 Current	3.5	4.0	Milliamperes
Plate Input	45	67.5	Watts
Screen Input	2.0	2.0	Watts
Plate Dissipation	13.3	16.7	Watts
Peak Heater-Cathode Voltage			
Heater Positive with Respect to Cathode	135	135	Volts
Heater Negative with Respect to Cathode	135	135	Volts
Grid-Number 1 Circuit Resistance 	30000	30000	Ohms
Bulb Temperature at Hottest Point	220	220	C

TYPICAL OPERATION

DC Plate Voltage	400	475	600	Volts
DC Screen Voltage††	150	135	150	Volts
From a Series Resistor of	33000	51000	56000	Ohms
DC Grid-Number 1 Voltage§§	-87	-77	-87	Volts
From a Grid Resistor of	27000	27000	27000	Ohms
Peak RF Grid-Number 1 Voltage	107	95	107	Volts
DC Plate Current	112	94	112	Milliamperes
DC Screen Current	7.8	6.4	7.8	Milliamperes
DC Grid-Number 1 Current, approximate	3.4	2.8	3.4	Milliamperes
Driving Power, approximate	0.4	0.3	0.4	Watts
Power Output, approximate	32	34	52	Watts

RADIO-FREQUENCY POWER AMPLIFIER AND OSCILLATOR—CLASS C TELEGRAPHY ¶¶ AND RADIO-FREQUENCY POWER AMPLIFIER—CLASS C FM TELEPHONY

MAXIMUM RATINGS, ABSOLUTE VALUES

	CCS†	ICAS§	
DC Plate Voltage	600	750	Volts
DC Screen Voltage	250	250	Volts
DC Grid-Number 1 Voltage	—150	—150	Volts
DC Plate Current	140	150	Milliamperes
DC Grid-Number 1 Current	3.5	4.0	Milliamperes
Plate Input	67.5	90	Watts
Screen Input	3.0	3.0	Watts
Plate Dissipation	20	25	Watts
Peak Heater-Cathode Voltage			
Heater Positive with Respect to Cathode	135	135	Volts
Heater Negative with Respect to Cathode	135	135	Volts
Grid-Number 1 Circuit Resistance 	30000	30000	Ohms
Bulb Temperature at Hottest Point	220	220	C

TYPICAL OPERATION AS AMPLIFIER UP TO 60 MEGACYCLES

DC Plate Voltage	500	600	600	750	Volts
DC Screen Voltage # #	170	150	180	160	Volts
From a Series Resistor of	36000	51000	43000	56000	Ohms
DC Grid-Number 1 Voltage ¶¶	—66	—58	—71	—62	Volts
From a Grid Resistor of	27000	20000	24000	20000	Ohms
From a Cathode Resistor of	470	470	430	470	Ohms
Peak RF Grid-Number 1 Voltage	84	73	91	79	Volts
DC Plate Current	135	112	150	120	Milliamperes
DC Screen Current	9.0	9.0	10	11	Milliamperes
DC Grid-Number 1 Current, approximate	2.5	2.8	2.8	3.1	Milliamperes
Driving Power, approximate	0.2	0.2	0.3	0.2	Watts
Power Output, approximate	48	52	66	70	Watts

TYPICAL OPERATION AS AMPLIFIER AT 175 MEGACYCLES

DC Plate Voltage	320	400	Volts
DC Screen Voltage # #	180	190	Volts
From a Series Resistor of	13000	20000	Ohms
DC Grid-Number 1 Voltage ¶¶	—51	—54	Volts
From a Grid Resistor of	27000	24000	Ohms
From a Cathode Resistor of	330	330	Ohms
Peak RF Grid-Number 1 Voltage	64	68	Volts
DC Plate Current	140	150	Milliamperes
DC Screen Current	10	10.4	Milliamperes
DC Grid-Number 1 Current, approximate	2.0	2.2	Milliamperes
Driving Power, approximate	3.0	3.0	Watts
Power Output, approximate	25	35	Watts

MAXIMUM RATINGS, OPERATING FREQUENCY

The 6146 can be operated under Class C conditions at frequencies as high as 60 megacycles. The tube may be operated at higher frequencies provided the maximum values of plate voltage and power input are reduced as the frequency is raised (other maximum ratings are the same as shown above). The tabulation below shows the highest percentage of maximum plate voltage and power input that can be used up to 175 megacycles for the various classes of service.

Frequency	60	80	125	150	160	175	Megacycles
Percentage of Maximum Plate Voltage and Plate Input							
Class C, Plate Modulated or Unmodulated							
Voltage	100	84	65	58	56	53	Percent
Input	100	92	78	72	70	67	Percent

INITIAL CHARACTERISTICS LIMITS

	Minimum	Maximum	
Heater Current			
Ef = 6.3 volts	1.175	1.325	Amperes
Plate Current			
Ef = 6.3 volts, Eb = 300 volts	46	94	Milliamperes
Ec2 = 200 volts, Ec1 = -33 volts			
Screen Current			
Ef = 6.3 volts, Eb = 300 volts		5.5	Milliamperes
Ec1 = -33 volts			
Power Output			
Ef = 6.3 volts, Eb = 600 volts, Ec2 = 180 volts, Rg1 = 30000 ± 10% ohms, Ib = 100 to 112 ma, Icl = 2 to 2.5 ma. Measured in self-excited oscillator circuit operating at 15 mc.	47		Watts
Interelectrode Capacitances			
Grid-Number 1 to Plate (g1 to p)		0.24	μμf
Input (g1 to h, k, g2, g3, IS, and base sleeve)	11.1	15.9	μμf
Output (p to h, k, g2, g3, IS, and base sleeve)	6.4	10.6	μμf
Measured without external shield.			

* Without external shield.

† Subscript 1 indicates that grid-number 1 current does not flow during any part of the input cycle.

‡ Continuous Commercial Service.

§ Intermittent Commercial and Amateur Service.

|| Averaged over any audio-frequency cycle of sine-wave form.

¶ The driver stage should be capable of supplying the number 1 grids of the class AB₁ stage with the specified driving voltage at low distortion.

The type of input-coupling network used should not introduce too much resistance in the grid-number 1 circuit. Transformer or impedance coupling devices are recommended. When grid-number 1 is operated in the negative region with fixed bias, the d-c grid-number 1 circuit resistance should not exceed the specified value of 0.1 megohm. For higher values of d-c grid-number 1 circuit resistance, cathode bias is required. Under no circumstances should the total d-c grid-number 1 circuit resistance exceed the specified value of 0.5 megohm. Cathode-bias operation is not recommended for AF Power Amplifier and Modulator—Class AB₁ or AB₂.

¢ Preferably obtained from a separate source or from the plate-voltage supply with a voltage divider.

**Subscript 2 indicates that grid-number 1 current flows during some part of the input cycle.

†† Driver stage should be capable of supplying the specified driving power at low distortion to number 1 grids of the AB₂ stage. To minimize distortion, the effective resistance per grid-number 1 circuit of the AB₂ stage should be held at a low value. For this purpose, the use of transformer coupling is recommended. In no case, however, should the total d-c grid-number 1 circuit resistance exceed 30000 ohms when the 6146 is operated at maximum ratings. For operation at less than maximum ratings, the d-c grid-number 1 circuit resistance may be as high as 100000 ohms.

‡‡ Obtained preferably from a separate source modulated with the plate supply, or from the modulated plate supply through a series resistor.

§§ Obtained from grid-number 1 resistor or from a combination of grid-number 1 resistor with either fixed supply or cathode resistor.

||| When grid-number 1 is driven positive and the 6146 is operated at maximum ratings, the total d-c grid-number 1 circuit resistance should not exceed the specified value of 30000 ohms. If this value is insufficient to provide adequate bias, the additional required bias must be supplied by a cathode resistor or fixed supply. For operation at less than maximum ratings, the d-c grid-number 1 circuit resistance may be as high as 100000 ohms.

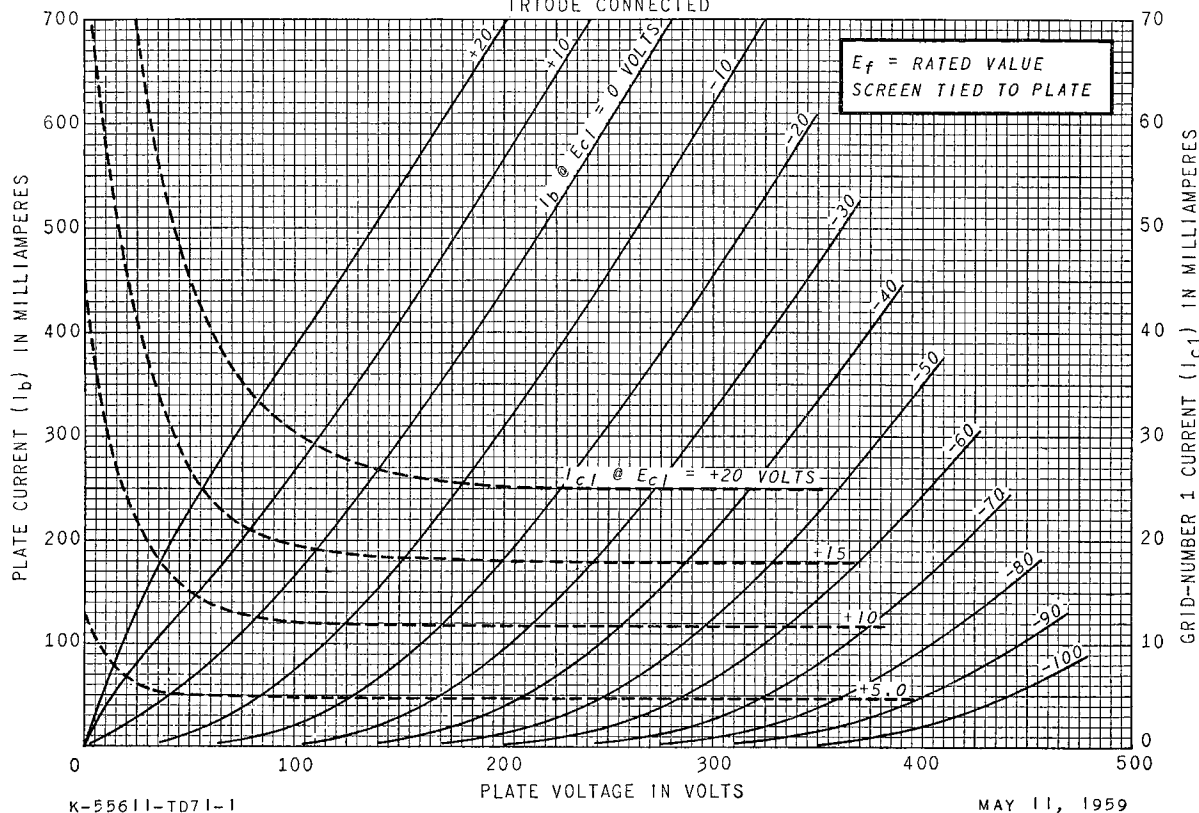
¶¶ Key-down conditions per tube without amplitude modulation. Amplitude modulation essentially negative may be used if the positive peak of the audio-frequency envelope does not exceed 115 percent of the carrier conditions.

Obtained preferably from a separate source, or from the plate-supply voltage with a voltage divider, or through a series resistor. A series grid-number 2 resistor should be used only when the 6146 is used in a circuit which is not keyed. Grid-number 2 voltage must not exceed 400 volts under key-up conditions.

¢¢ Obtained from fixed supply by grid-number 1 resistor, by cathode resistor, or by combination methods.

AVERAGE PLATE CHARACTERISTICS

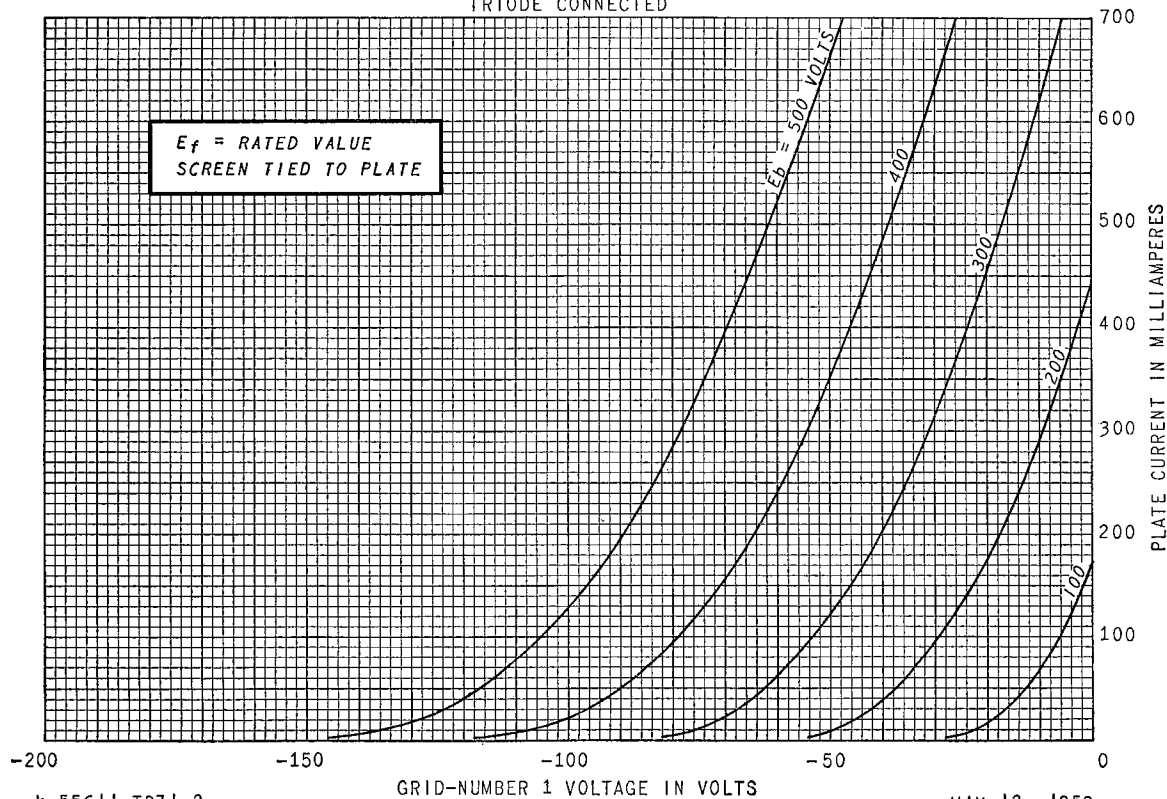
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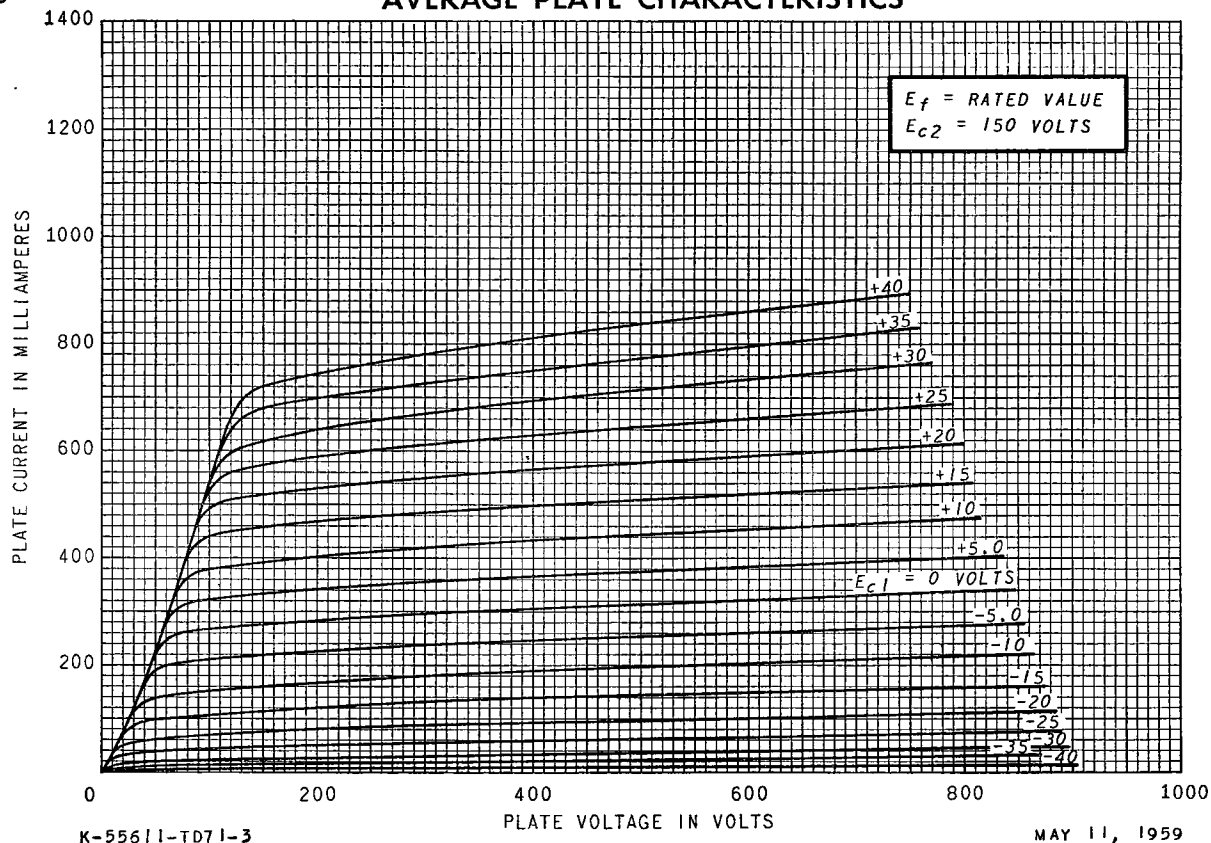
AVERAGE TRANSFER CHARACTERISTICS

TRIODE CONNECTED

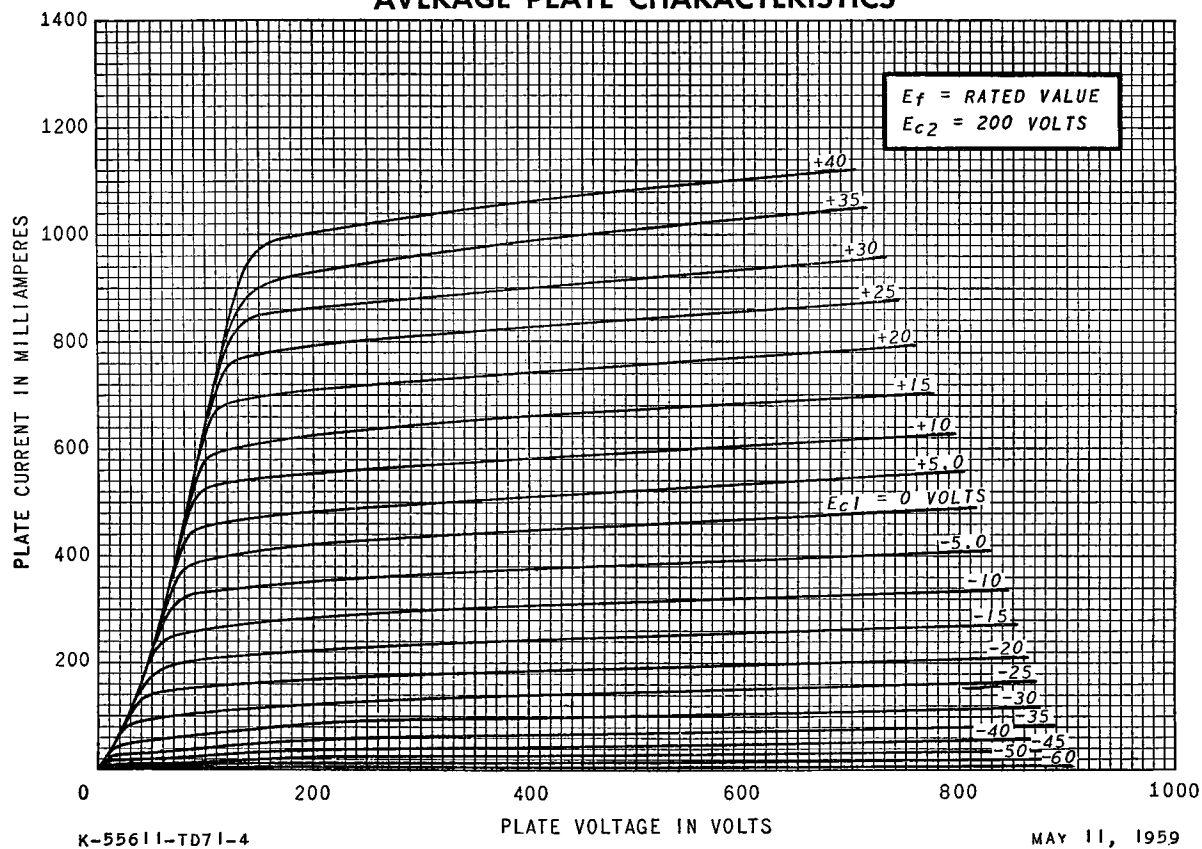


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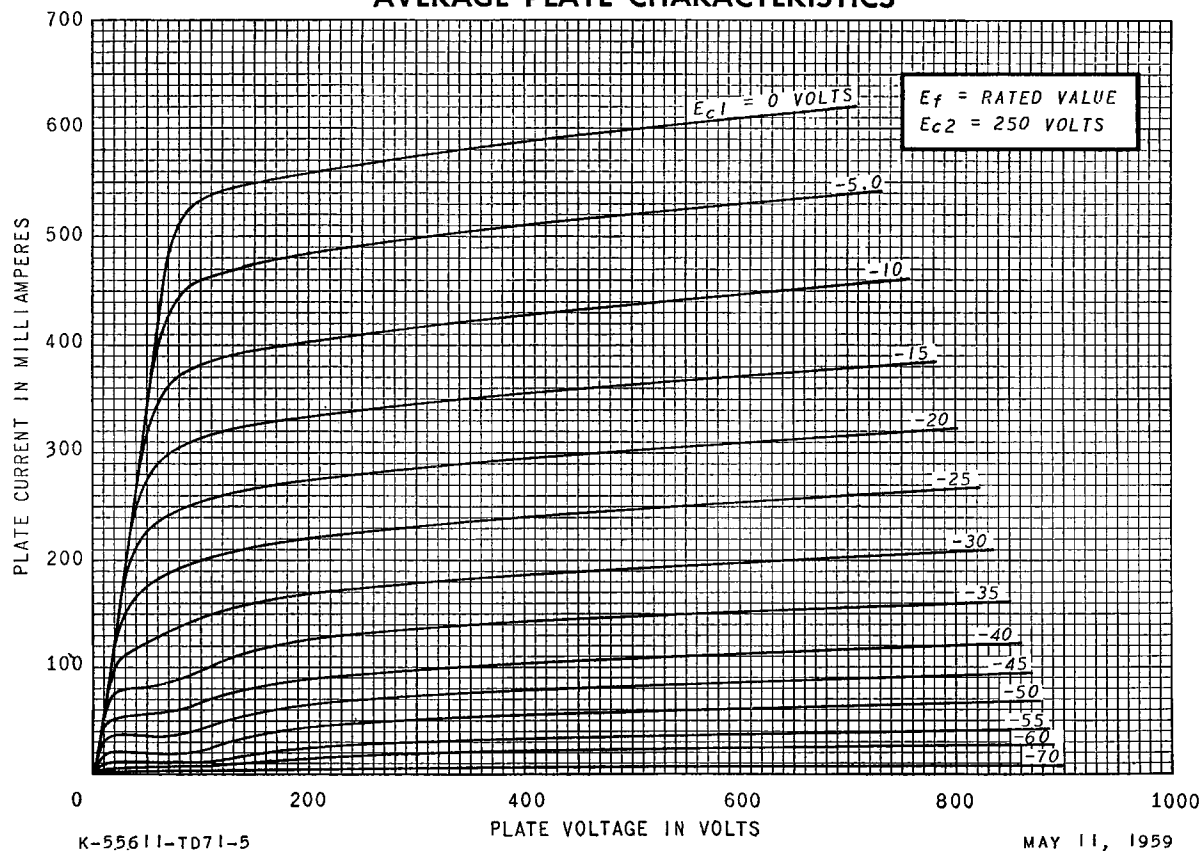
AVERAGE PLATE CHARACTERISTICS



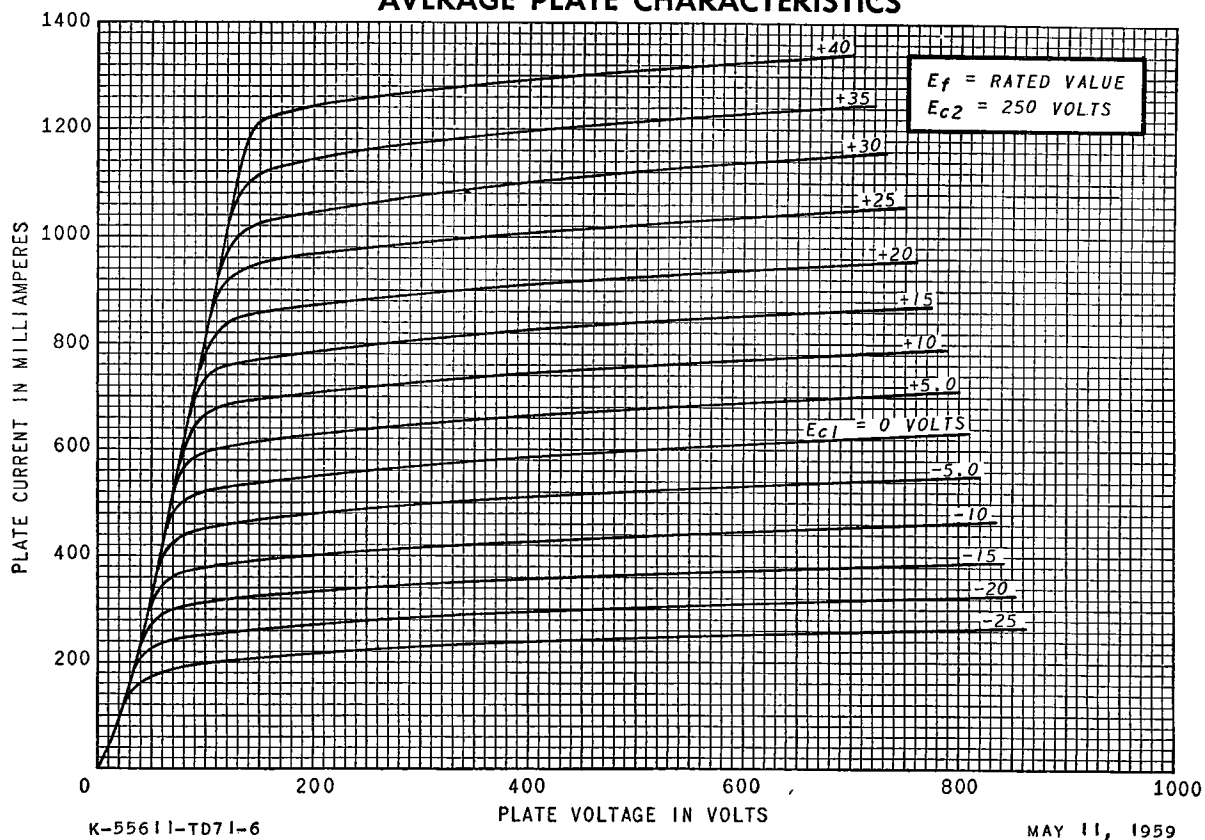
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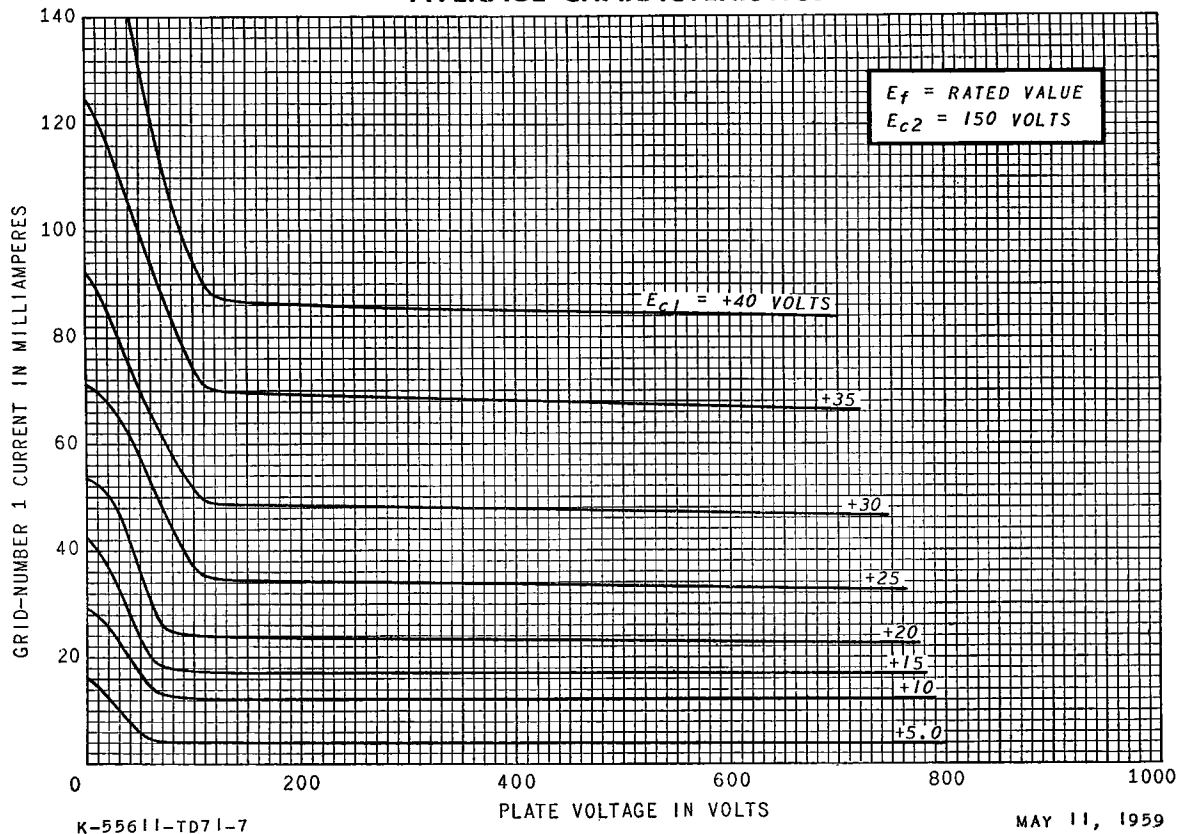
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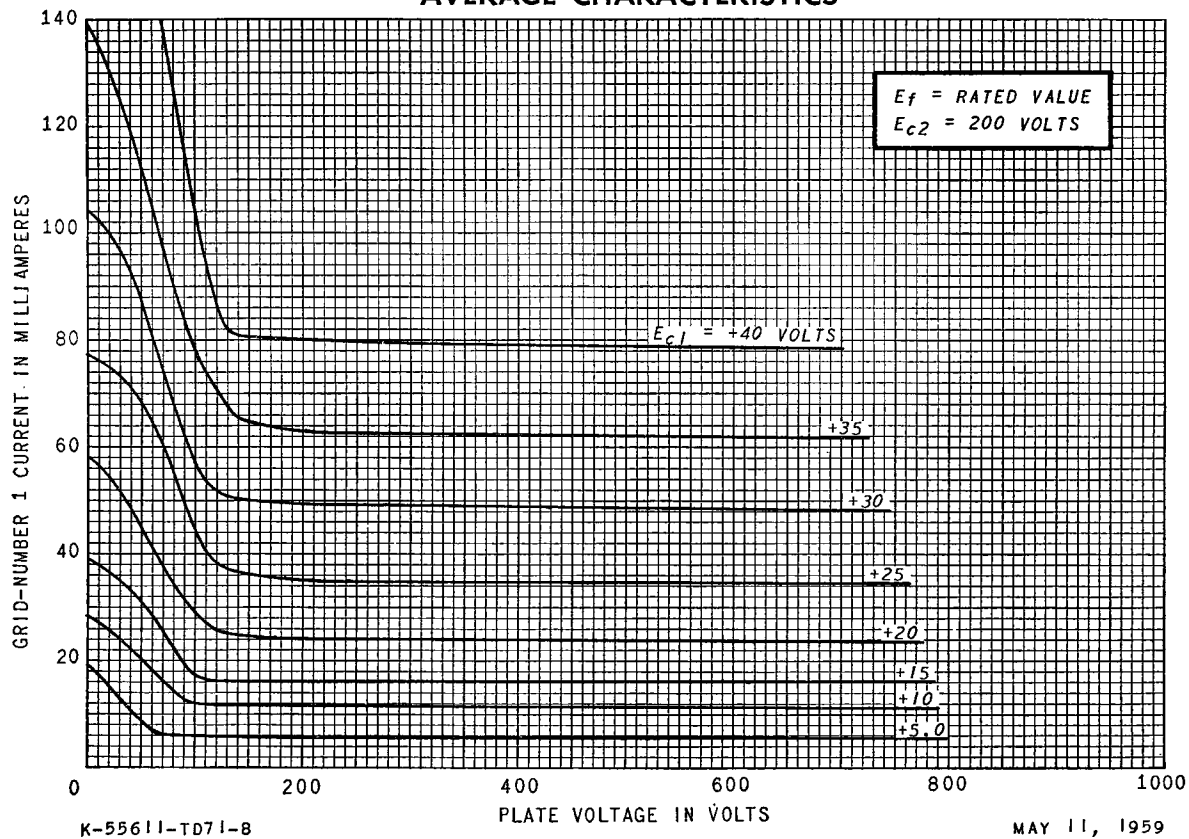
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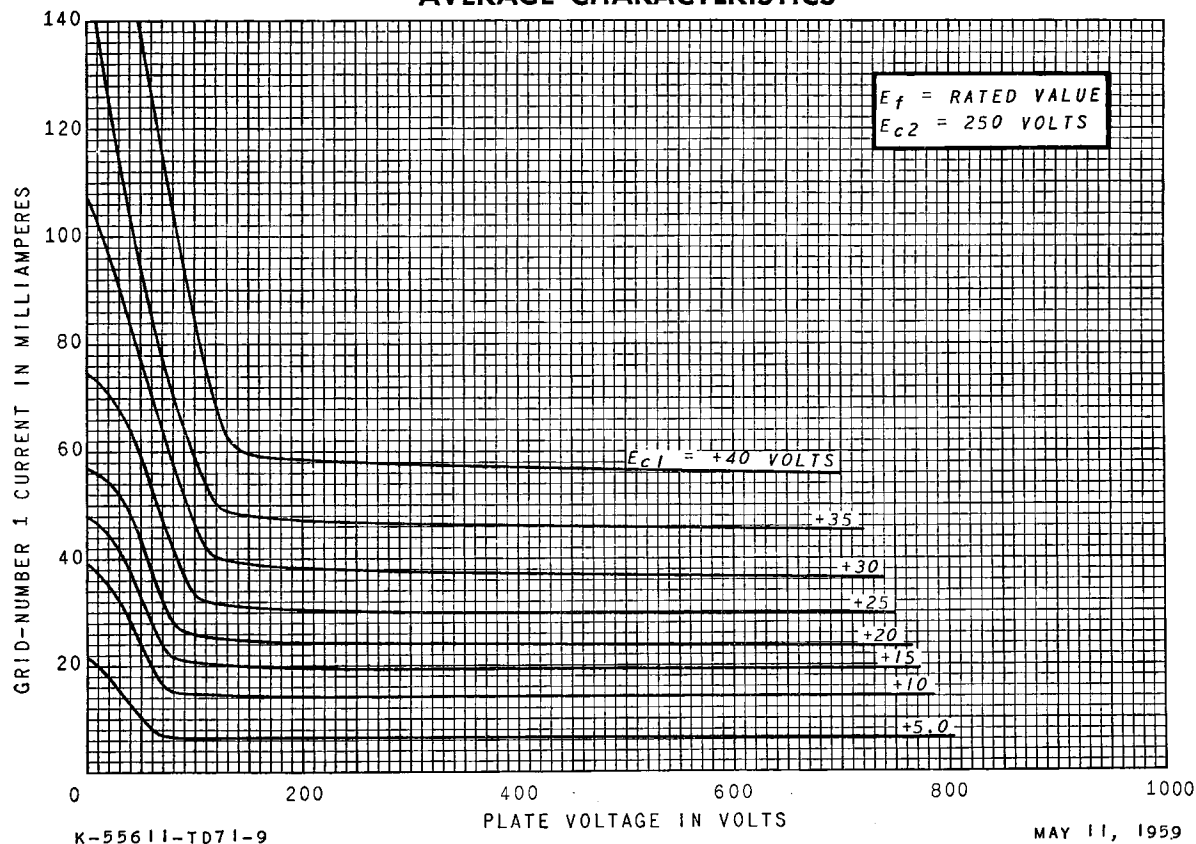
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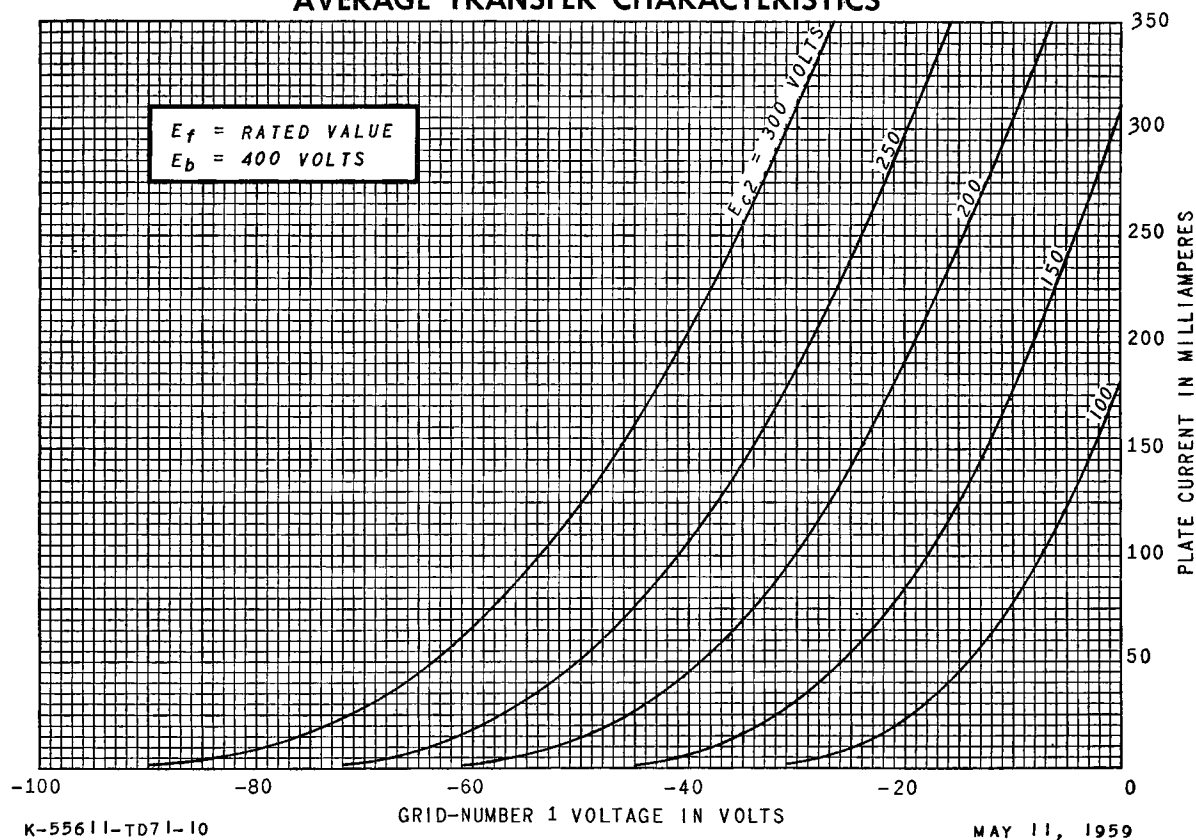
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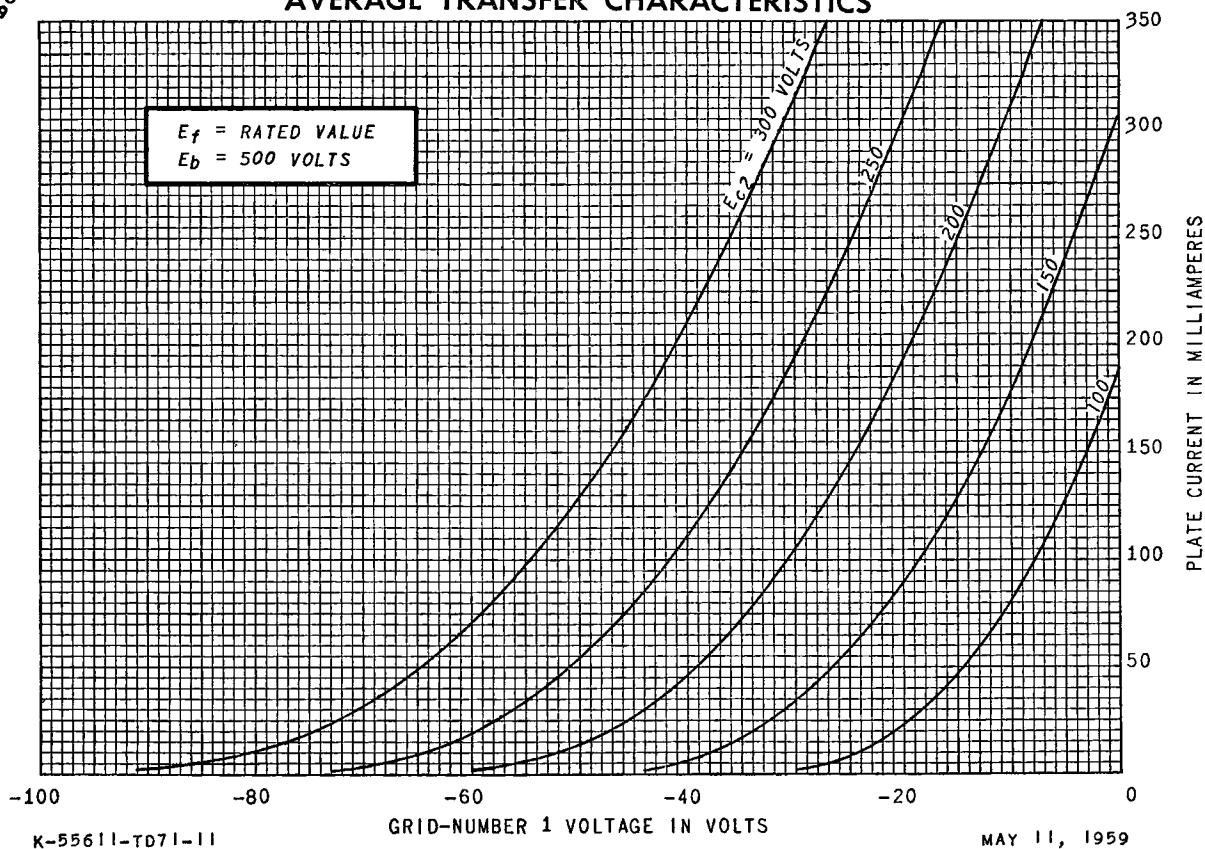
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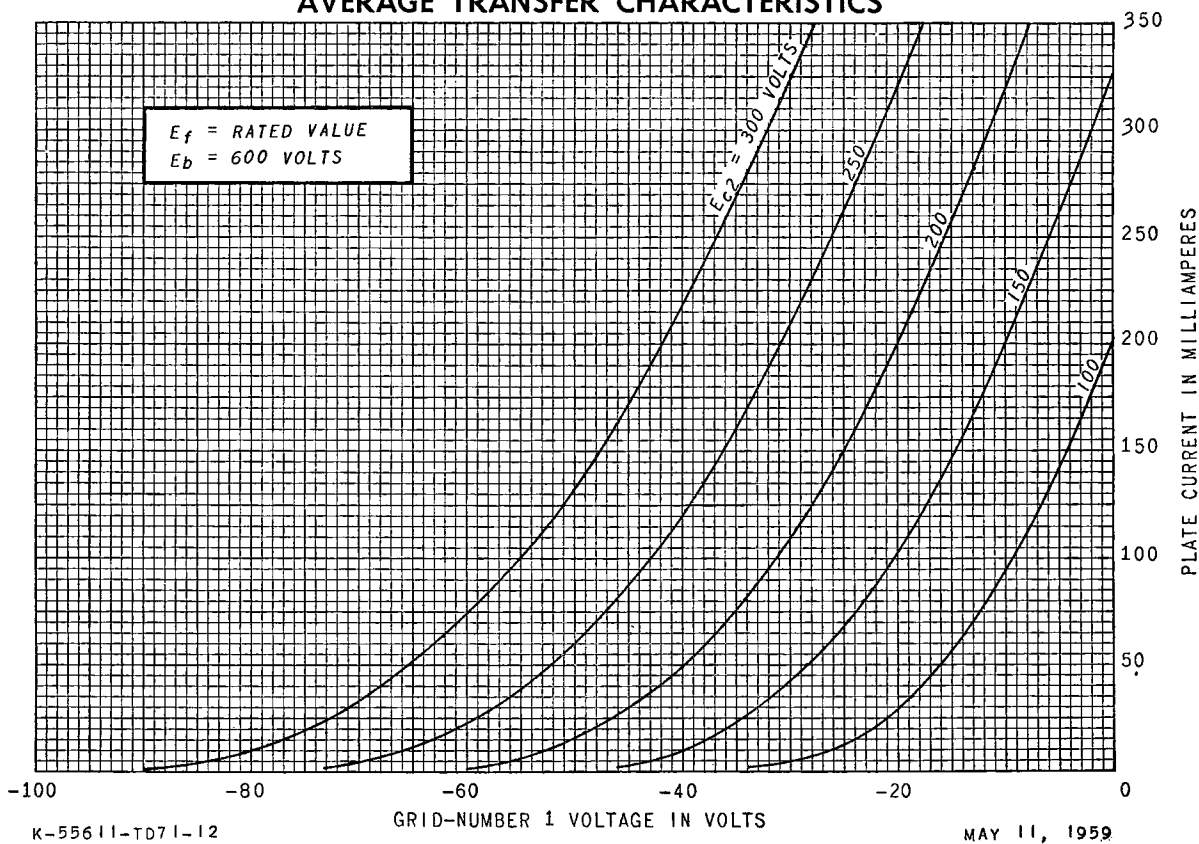
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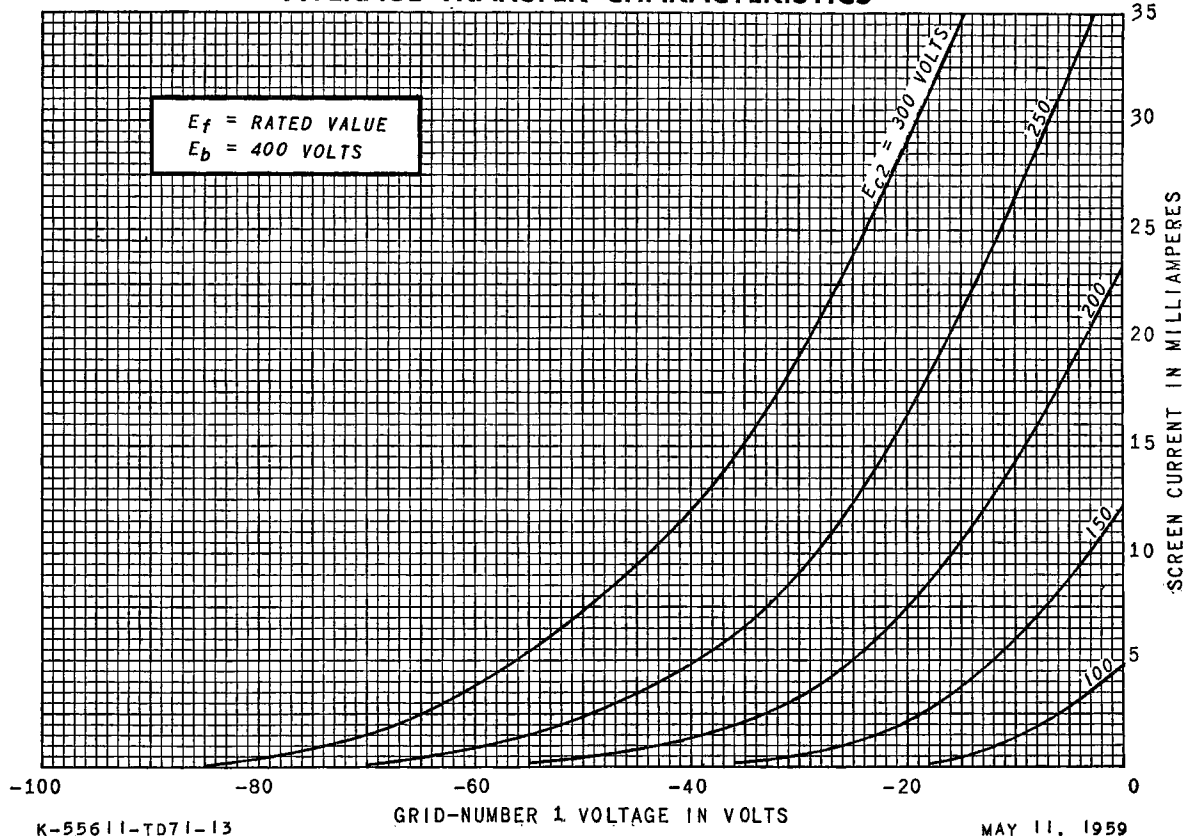
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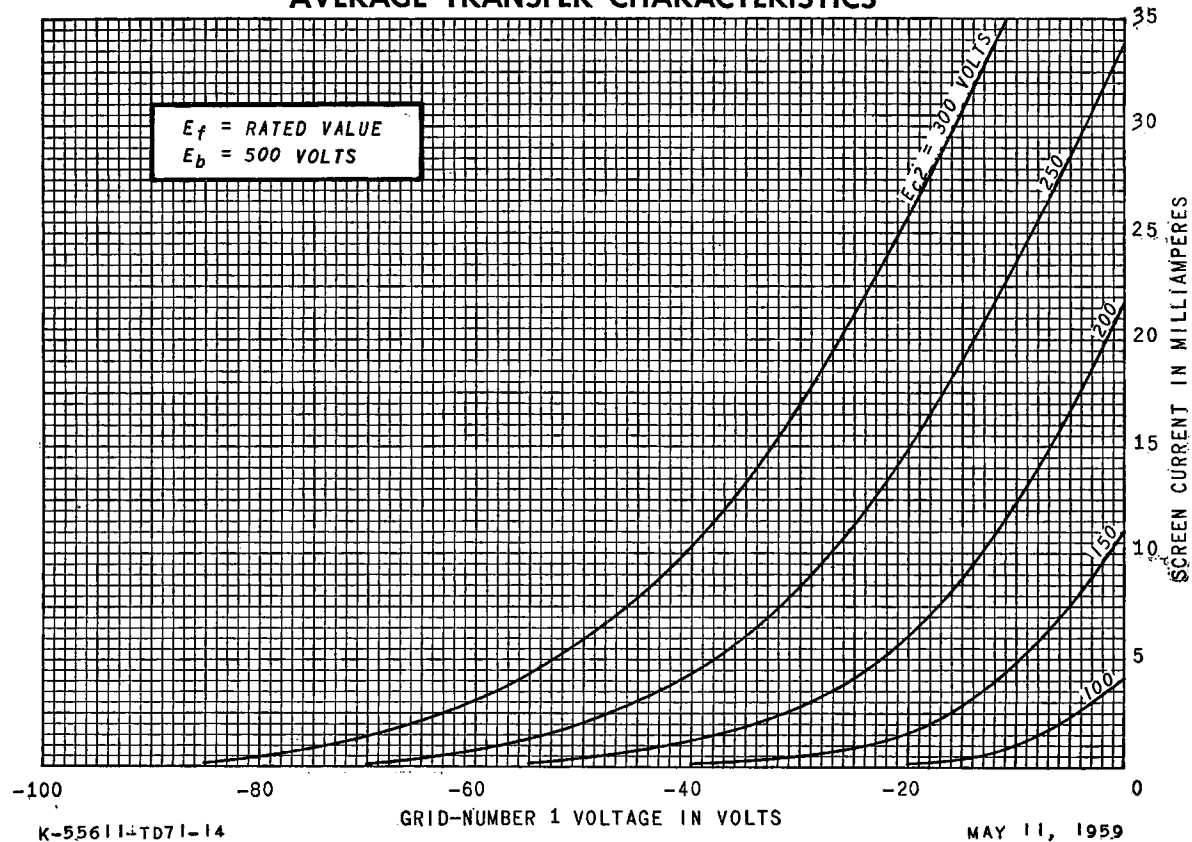
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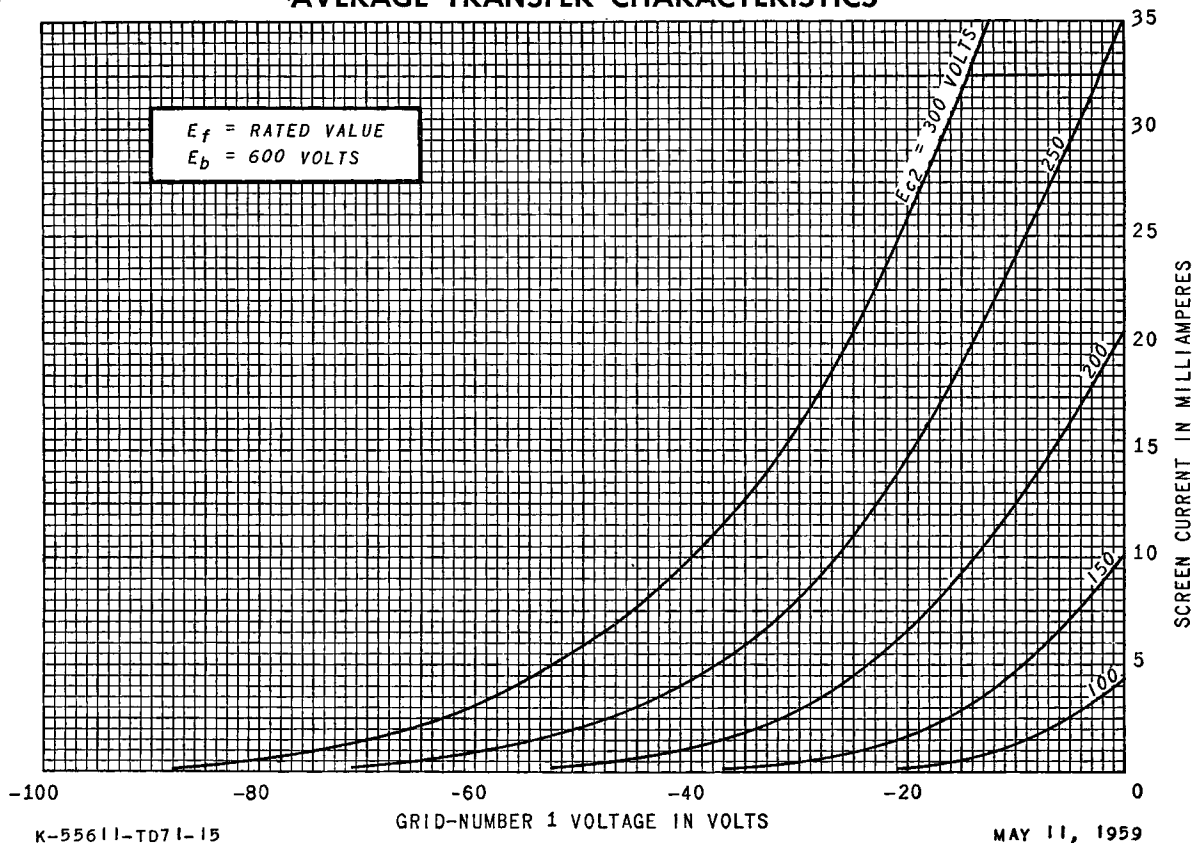
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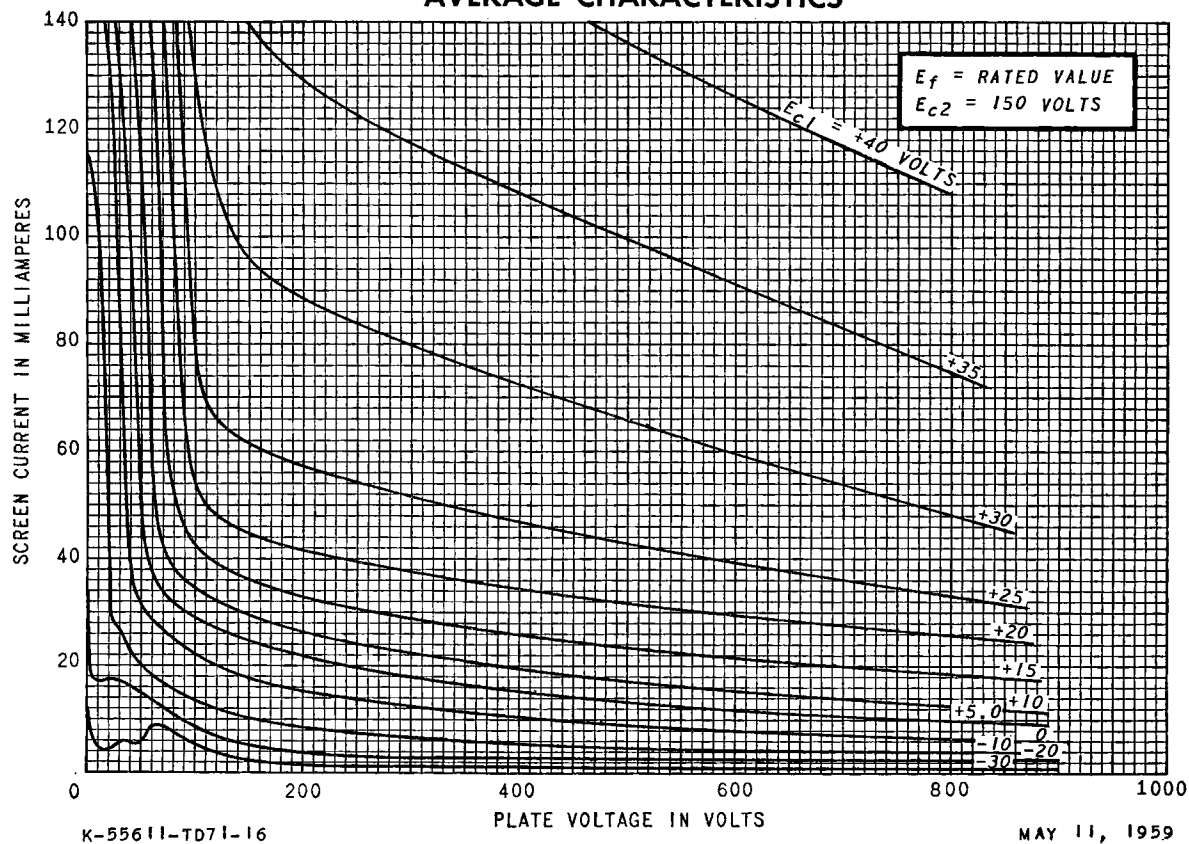
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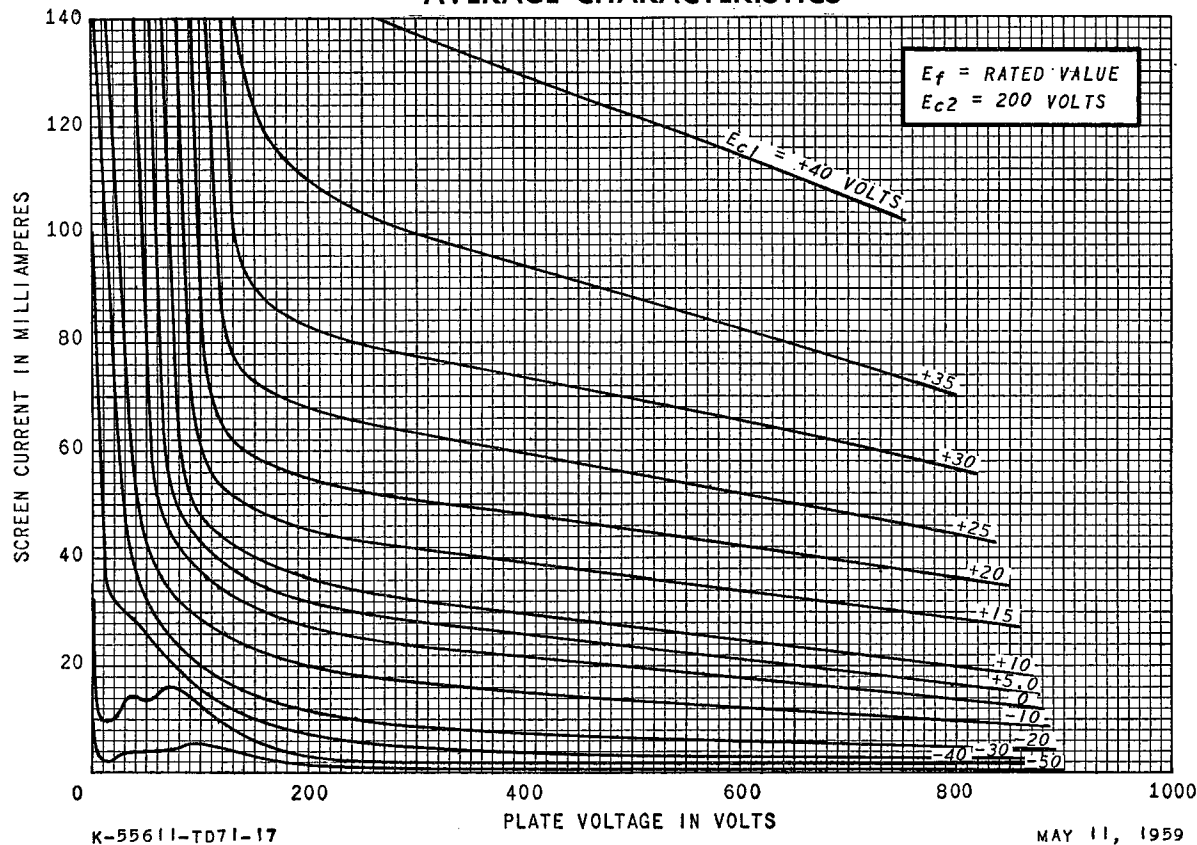
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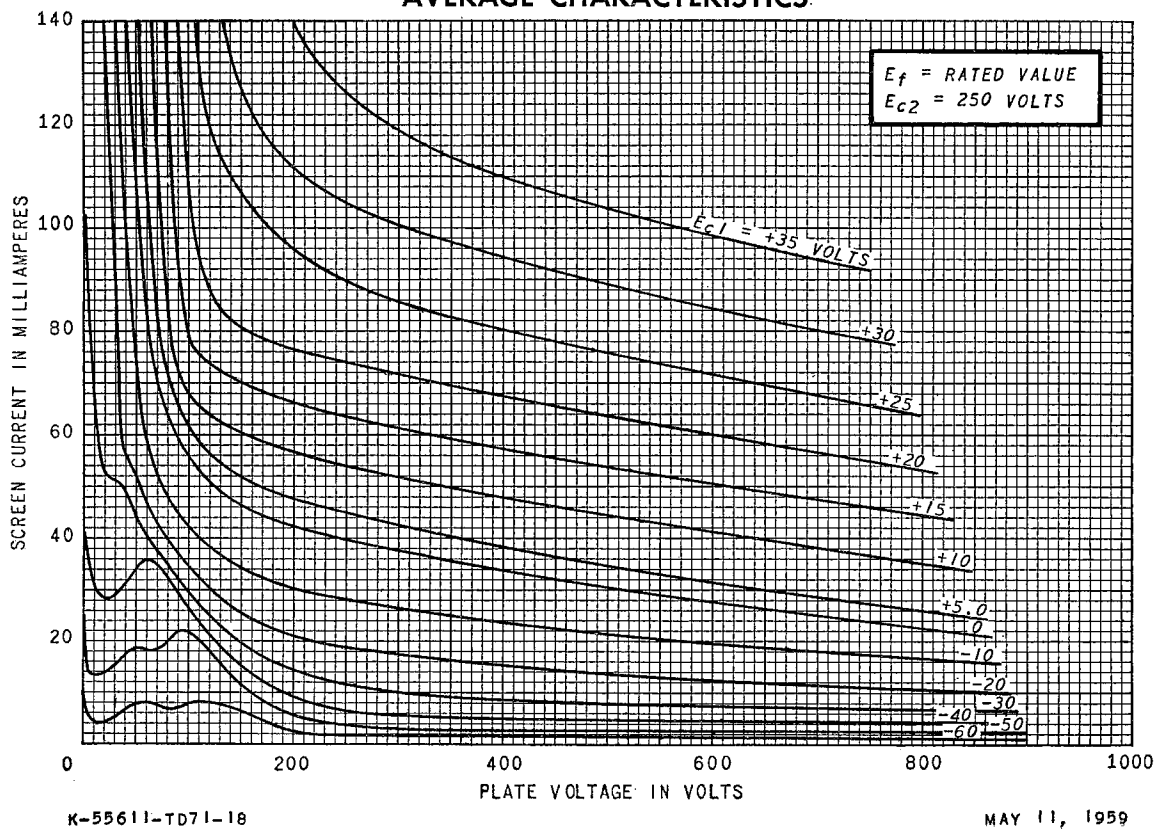
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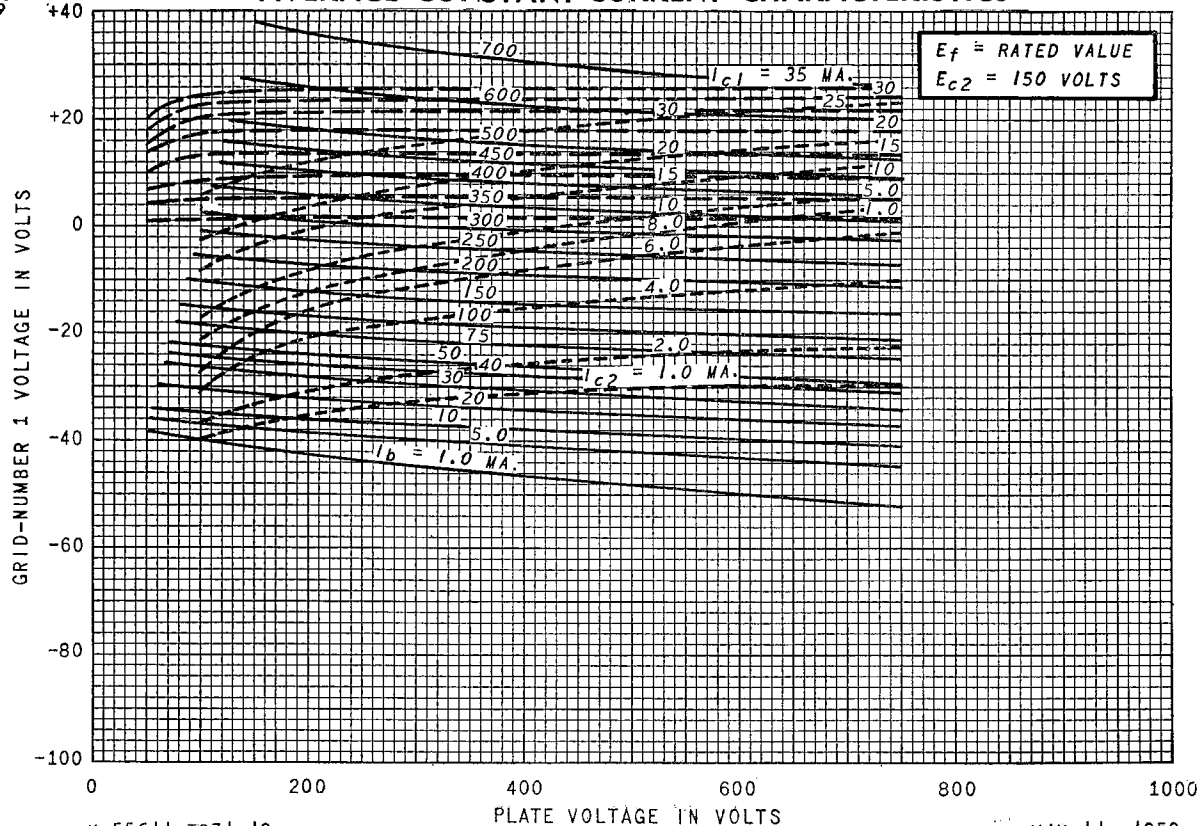
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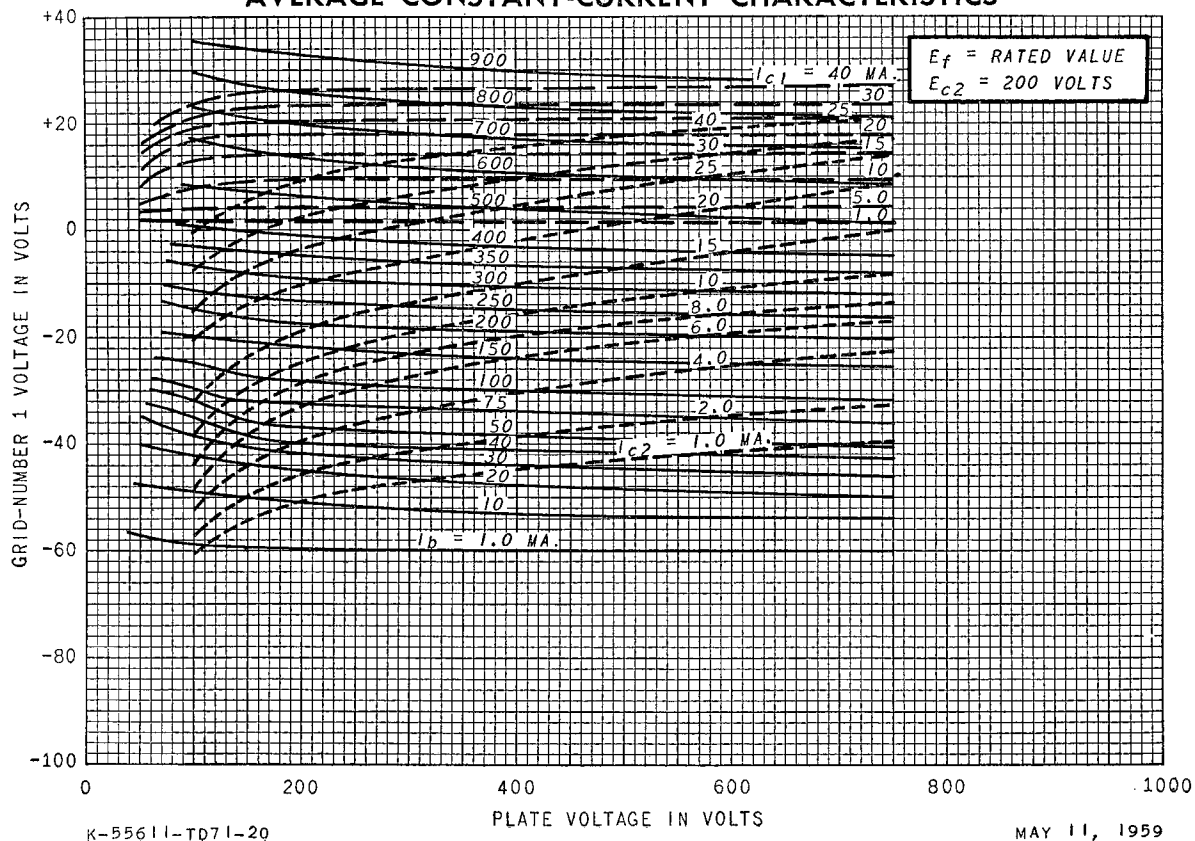
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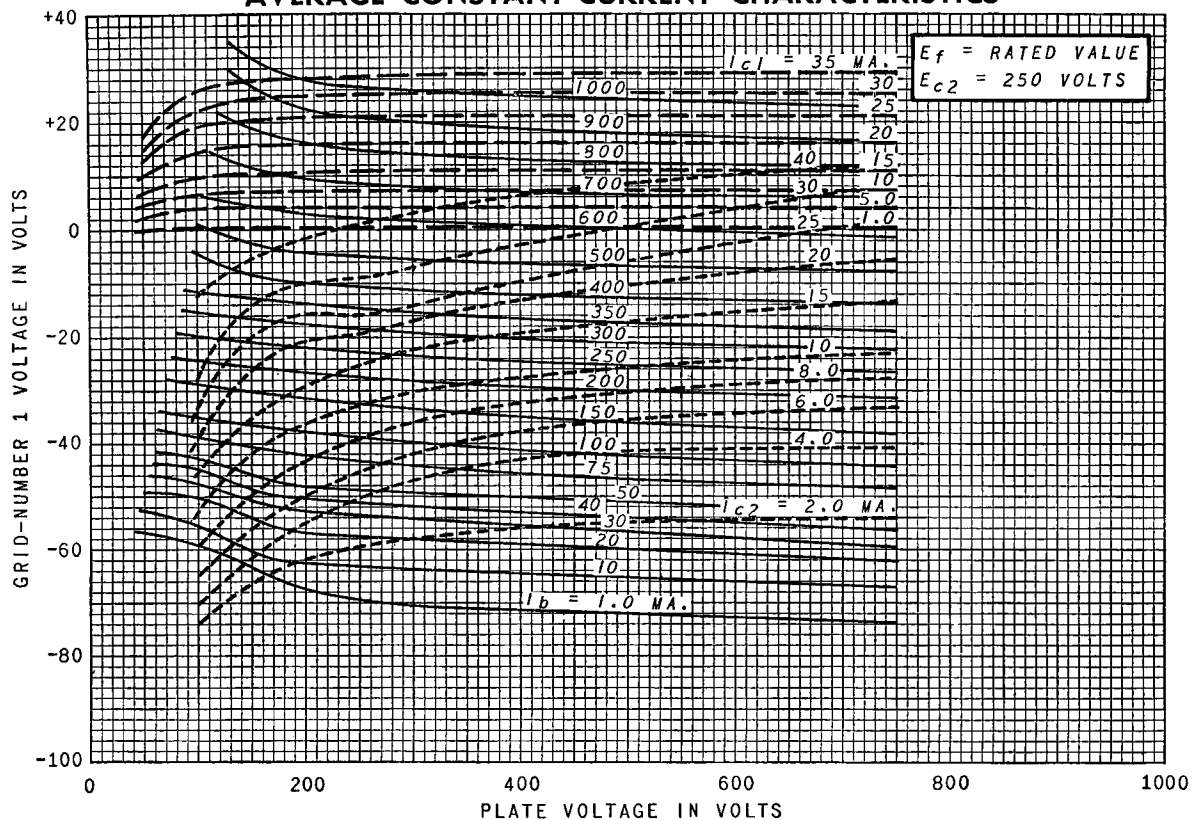
AVERAGE CONSTANT-CURRENT CHARACTERISTICS



AVERAGE CONSTANT-CURRENT CHARACTERISTICS



AVERAGE CONSTANT-CURRENT CHARACTERISTICS



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